# ModelAudit — Summary (page 1)


```
{
  "dataset": "german",
  "model": "scorecard",
  "scope": "research demonstration only; thresholds are local review heuristics",
  "evidence_sha256": "f9753a934bea3ee8b3f225b0638ef70bc2ee1f6e58e3916d46e66f587c8b156a",
  "findings": [
    {
      "test_id": "CAL-ECE-01",
      "severity": "medium",
      "confirmation": "reproduced"
    },
    {
      "test_id": "STAB-CSI-01",
      "severity": "medium",
      "confirmation": "reproduced"
    },
    {
      "test_id": "FAIR-SEX-01",
      "severity": "medium",
      "confirmation": "reproduced"
    },
    {
      "test_id": "FAIR-AGE-01",
      "severity": "high",
      "confirmation": "reproduced"
    },
    {
      "test_id": "ROB-MONO-01",
      "severity": "high",
      "confirmation": "reproduced"
    }
  ]
}
```

# ModelAudit — PERF-AUC-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "auc": 0.8141714285714287,
    "auc_ci95": [
      0.7629245274824543,
      0.8668489520696215
    ],
    "ks": 0.5295238095238095,
    "gini": 0.6283428571428573,
    "count": 250
  },
  "threshold": {
    "metric": "auc",
    "warn_below": 0.75,
    "fail_below": 0.7
  }
}
```

# ModelAudit — CAL-ECE-01 (page 1)


```
{
  "status": "warn",
  "values": {
    "brier": 0.15648754620485295,
    "ece": 0.04711499223229103,
    "calibration_in_the_large": -0.005826874641605528,
    "hosmer_lemeshow_stat": 6.19388928599065,
    "hosmer_lemeshow_df": 8,
    "hosmer_lemeshow_p": 0.6255233912183731,
    "reliability_bins": [
      {
        "bin": 0,
        "count": 25,
        "observed_rate": 0.04,
        "predicted_rate": 0.03191449935790145
      },
      {
        "bin": 1,
        "count": 25,
        "observed_rate": 0.04,
        "predicted_rate": 0.07453117241587133
      },
      {
        "bin": 2,
        "count": 25,
        "observed_rate": 0.0,
        "predicted_rate": 0.12501639650584248
      },
      {
        "bin": 3,
        "count": 25,
        "observed_rate": 0.2,
        "predicted_rate": 0.1644679628625908
      },
      {
        "bin": 4,
        "count": 25,
        "observed_rate": 0.2,
        "predicted_rate": 0.2189447998877008
      },
      {
        "bin": 5,
        "count": 25,
        "observed_rate": 0.28,
        "predicted_rate": 0.2924595594061095
      },
      {
        "bin": 6,
        "count": 25,
        "observed_rate": 0.4,
        "predicted_rate": 0.3874203219272815
      },
      {
        "bin": 7,
        "count": 25,
        "observed_rate": 0.56,
```

```
        "predicted_rate": 0.4766219149204106
      },
      {
        "bin": 8,
        "count": 25,
        "observed_rate": 0.64,
        "predicted_rate": 0.573134712978388
      },
      {
        "bin": 9,
        "count": 25,
        "observed_rate": 0.64,
        "predicted_rate": 0.7137574061539584
      }
    ]
  },
  "threshold": {
    "metric": "ece",
    "warn_above": 0.03,
    "fail_above": 0.05
  }
}
```

# ModelAudit — STAB-PSI-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "psi": 0.03466715595663763,
    "baseline_test_psi": 0.05271587982764225,
    "scenario": "amount +25% and age +5"
  },
  "threshold": {
    "metric": "psi",
    "warn_above": 0.1,
    "fail_above": 0.25
  }
}
```

# ModelAudit — STAB-CSI-01 (page 1)


```
{
  "status": "warn",
  "values": {
    "max_csi": 0.12116270428342113,
    "feature": "A05",
    "baseline_test_csi": 0.05001226836672981
  },
  "threshold": {
    "metric": "max_csi",
    "warn_above": 0.1,
    "fail_above": 0.25
  }
}
```

# ModelAudit — FAIR-SEX-01 (page 1)


```
{
  "status": "warn",
  "values": {
    "threshold": 0.25,
    "groups": {
      "female": {
        "approval_rate": 0.4457831325301205,
        "good_outcome_tpr": 0.6415094339622641
      },
      "male_or_other": {
        "approval_rate": 0.5389221556886228,
        "good_outcome_tpr": 0.6639344262295082
      }
    },
    "demographic_parity_difference": 0.09313902315850231,
    "equalized_odds_difference": 0.1,
    "approval_rate_ratio": 0.8271753681392235,
    "overall_approval_rate": 0.508,
    "approved_observed_default_rate": 0.09448818897637795
  },
  "threshold": {
    "metric": "equalized_odds_difference",
    "warn_above": 0.05,
    "fail_above": 0.1
  }
}
```

# ModelAudit — FAIR-AGE-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "threshold": 0.25,
    "groups": {
      "35plus": {
        "approval_rate": 0.6391752577319587,
        "good_outcome_tpr": 0.7763157894736842
      },
      "under35": {
        "approval_rate": 0.42483660130718953,
        "good_outcome_tpr": 0.5656565656565656
      }
    },
    "demographic_parity_difference": 0.21433865642476918,
    "equalized_odds_difference": 0.21065922381711855,
    "approval_rate_ratio": 0.664663714948345,
    "overall_approval_rate": 0.508,
    "approved_observed_default_rate": 0.09448818897637795
  },
  "threshold": {
    "metric": "equalized_odds_difference",
    "warn_above": 0.05,
    "fail_above": 0.1
  }
}
```

# ModelAudit — ROB-MONO-01 (page 1)


```
{
  "status": "fail",
  "values": {
    "test_feature": "A02 loan duration months (stress proxy)",
    "violations": 21,
    "sample_rows": 250,
    "max_absolute_change": 0.11457027816975857,
    "missing_predictions_finite": true
  },
  "threshold": {
    "metric": "violations",
    "warn_above": 0,
    "fail_above": 1
  }
}
```

# ModelAudit — DQ-CAT-01 (page 1)


```
{
  "status": "pass",
  "values": {
    "rows": 250,
    "null_cells": 0,
    "duplicate_feature_rows": 0,
    "invalid_categories": 0,
    "label_out_of_range": 0,
    "id_and_target_excluded": true
  },
  "threshold": {
    "metric": "invalid_categories",
    "warn_above": 0,
    "fail_above": 100
  }
}
```

# ModelAudit — FAIR-TRADEOFF (page 1)


```
{
  "status": "information",
  "values": {
    "thresholds": [
      {
        "threshold": 0.15,
        "groups": {
          "female": {
            "approval_rate": 0.2289156626506024,
            "good_outcome_tpr": 0.3584905660377358
          },
          "male_or_other": {
            "approval_rate": 0.3413173652694611,
            "good_outcome_tpr": 0.45081967213114754
          }
        },
        "demographic_parity_difference": 0.1124017026188587,
        "equalized_odds_difference": 0.09232910609341172,
        "approval_rate_ratio": 0.6706827309236947,
        "overall_approval_rate": 0.304,
        "approved_observed_default_rate": 0.02631578947368421
      },
      {
        "threshold": 0.2,
        "groups": {
          "female": {
            "approval_rate": 0.3493975903614458,
            "good_outcome_tpr": 0.49056603773584906
          },
          "male_or_other": {
            "approval_rate": 0.4491017964071856,
            "good_outcome_tpr": 0.5655737704918032
          }
        },
        "demographic_parity_difference": 0.09970420604573982,
        "equalized_odds_difference": 0.07500773275595418,
        "approval_rate_ratio": 0.777991967871486,
        "overall_approval_rate": 0.416,
        "approved_observed_default_rate": 0.08653846153846154
      },
      {
        "threshold": 0.25,
        "groups": {
          "female": {
            "approval_rate": 0.4457831325301205,
            "good_outcome_tpr": 0.6415094339622641
          },
          "male_or_other": {
            "approval_rate": 0.5389221556886228,
            "good_outcome_tpr": 0.6639344262295082
          }
        },
        "demographic_parity_difference": 0.09313902315850231,
        "equalized_odds_difference": 0.1,
        "approval_rate_ratio": 0.8271753681392235,
        "overall_approval_rate": 0.508,
```

# ModelAudit — FAIR-TRADEOFF (page 2)


```
        "approved_observed_default_rate": 0.09448818897637795
      },
      {
        "threshold": 0.3,
        "groups": {
          "female": {
            "approval_rate": 0.5180722891566265,
            "good_outcome_tpr": 0.7358490566037735
          },
          "male_or_other": {
            "approval_rate": 0.592814371257485,
            "good_outcome_tpr": 0.7213114754098361
          }
        },
        "demographic_parity_difference": 0.07474208210085853,
        "equalized_odds_difference": 0.1111111111111111,
        "approval_rate_ratio": 0.8739199221126932,
        "overall_approval_rate": 0.568,
        "approved_observed_default_rate": 0.1056338028169014
      },
      {
        "threshold": 0.35,
        "groups": {
          "female": {
            "approval_rate": 0.5662650602409639,
            "good_outcome_tpr": 0.7735849056603774
          },
          "male_or_other": {
            "approval_rate": 0.6526946107784432,
            "good_outcome_tpr": 0.7704918032786885
          }
        },
        "demographic_parity_difference": 0.08642955053747925,
        "equalized_odds_difference": 0.1333333333333333,
        "approval_rate_ratio": 0.8675804133967061,
        "overall_approval_rate": 0.624,
        "approved_observed_default_rate": 0.1346153846153846
      }
    ]
  },
  "threshold": null
}
```

# ModelAudit — EXPL-SHAP-01 (page 1)


```
{
  "status": "information",
  "values": {
    "dataset": "german",
    "rows_explained": 100,
    "xgb_shap_global_top10": [
      {
        "feature": "cat__A01_A14",
        "mean_abs_shap": 0.4713290333747864
      },
      {
        "feature": "num__A02",
        "mean_abs_shap": 0.235748752951622
      },
      {
        "feature": "cat__A03_A34",
        "mean_abs_shap": 0.20800858736038208
      },
      {
        "feature": "num__A05",
        "mean_abs_shap": 0.19495922327041626
      },
      {
        "feature": "cat__A01_A11",
        "mean_abs_shap": 0.18921680748462677
      },
      {
        "feature": "num__A08",
        "mean_abs_shap": 0.1332424283027649
      },
      {
        "feature": "num__A13",
        "mean_abs_shap": 0.1268044263124466
      },
      {
        "feature": "cat__A04_A40",
        "mean_abs_shap": 0.11415299773216248
      },
      {
        "feature": "cat__A07_A74",
        "mean_abs_shap": 0.1027626022696495
      },
      {
        "feature": "cat__A04_A41",
        "mean_abs_shap": 0.10129211097955704
      }
    ],
    "xgb_first_case_adverse_top3": [
      {
        "feature": "cat__A01_A14",
        "shap_log_odds": 0.31417495012283325
      },
      {
        "feature": "num__A05",
        "shap_log_odds": 0.15592725574970245
      },
```

# ModelAudit — EXPL-SHAP-01 (page 2)


```
      {
        "feature": "num__A08",
        "shap_log_odds": 0.11538189649581909
      }
    ],
    "scorecard_first_case_reason_codes": [
      "A01",
      "A08",
      "A06"
    ],
    "scorecard_top3_overlap_under_1pct_amount_change": 1.0,
    "warning": "SHAP is model explanation in log-odds space, not a causal reason or lawful adverse-action
  notice."
  },
  "threshold": null
}
```